Programming for AI
Week 2 Activity

Task 1

Consider the C++ code from class below which takes a 3x3 matrix, finds values greater than 5 and stores them in a list.

```
#include <iostream>
#include <vector>

int main() {
    int matrix[3][3] = {{1, 6, 2}, {3, 8, 7}, {4, 5, 9}};
    std::vector<int> results;

    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            if (matrix[i][j] > 5) {
                results.push_back(matrix[i][j]);
            }
        }
    }
    // Result: {6, 8, 7, 9}
    return 0;
}
```

The Problem: Write a Python function **`extract_high_multiples(matrix)`** that accepts a 2-dimensional list (matrix) of arbitrary dimensions. The function must return a list of dictionaries.

1. Selection Criteria: An element should be included only if:
   - It is strictly greater than 5.
   - It is a multiple of 3.
2. Output Format: Each entry in the returned list must be a dictionary with the following keys:
   - 'row': The row index of the element.
   - 'col': The column index of the element.
   - 'val': The actual value of the element.
3. Constraint: You must implement the core logic using a single nested list comprehension (no multi-line for loops or .append() calls inside the function).

Task 2

The Goal: You have three lists: **cities** = ["Lahore", "Karachi", "Islamabad"], **temps** = [32, 35, 28], and **humidity** = [60, 70, 45].

The Problem: Write a Python function **`summerize(cities, temps, humidity)`** using list comprehension and zip(). The function must return a list of strings that says:

- “The weather in Lahore is 32C with 60% humidity.” (and so on for all cities).

Task 3

The Email Cleaner

You have a list of messy emails: `emails = [" ALI@gooGle.coM", "sara@Yahoo.com ", " AHMED@uCP.EdU.PK "]`

Problem: Write a Python function **`sanitize(emails)`** using a **list comprehension** to strip the whitespace and make them all lowercase. The function must return the sanitized list.

Task 4

Data Filtering

Given a list of temperatures in Celsius: `c_temps = [0, 12, 34, 100, -5, 25]`

Problem:

1. Use `filter` and a `lambda` to keep only positive temperatures. The lambda must stored in a variable and passed to the filter as a parameter.
2. Use `map` to convert the remaining positive temps to Fahrenheit (F=C×9/5+32). Use 2nd lambda function for mapping.

Task 5

The Dictionary Builder

Given two lists: `keys = ["ID", "Name", "Role"] values = [101, "Awais", "Faculty"]`

Problem: Write a Python function **`combine(keys, values)`** using a **dictionary comprehension** and `zip()` to combine them into one dictionary. The function must return the dictionary.